SAKSHI TAPARE

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SUMMARY

AI and Data Engineer with **nearly three years** building production systems where machine learning meets data infrastructure. Ships the full path from schema and ingestion through model deployment to the backend services that serve them, across ML pipelines, LLM and RAG applications, event driven automation and ETL on AWS.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, HTML and CSS
AI and ML	PyTorch, TensorFlow, Scikit Learn, Hugging Face, LangChain, MCP, RAG Systems, LLMs, Prompt Engineering, AI Agents, NLP, OpenCV
Data Engineering	ETL Pipelines, Pandas, NumPy, SQLAlchemy, Schema Design, Query Optimisation, ChromaDB, Vector Databases
Cloud and Automation	AWS, Lambda, API Gateway, n8n, FastAPI, REST APIs, WebSockets, Docker, MLOps, Git, Tally Integration
Visualisation	Power BI, Tableau, Plotly, Streamlit

EXPERIENCE

Senior Data Engineer / AI Engineer, K.B. Ajmera
Mumbai, India | Jul 2025 to Present

- Architected and deployed backend data pipelines sustaining **50,000+ records a day**, replacing manual inventory and invoice handling with **n8n** and Python automation.
- **Tripled query performance** by redesigning database schemas and deploying ML components on **AWS**.
- Cut manual query time **70%** with a **natural language SQL interface** built on intent detection and fuzzy entity resolution, now running in production.
- Built an **MCP server** over the inventory database, exposing **typed read only tools** to any Model Context Protocol client.
- Integrated the in house order system with **Tally**, eliminating duplicate data entry and keeping sales records and accounts reconciled automatically.
- Automated receivables follow up, tracking outstanding invoices and issuing scheduled **payment reminders** that removed the manual chasing cycle.
- Built a **bill of quantities generator** turning project inputs into itemised quantity and cost breakdowns, removing a repetitive estimation task.
- Shipped a **virtual try on** feature for a client fashion storefront, compositing garment imagery onto customer photographs so shoppers preview fit.

Executive Data Analyst, Pentagon Systems
Mumbai, India | Jan 2025 to Jun 2025

- Built end to end **ETL pipelines** automating extraction, transformation and load across heterogeneous source systems.
- Cut stakeholder reporting time **60%** with interactive **Power BI** dashboards built over the resulting data models.
- Delivered exploratory analysis and **predictive models** on **AWS** hosted analytics infrastructure.

Application Specialist, MIT Technology
Mumbai, India | Jan 2024 to Dec 2024

- Trained **100+ learners** in Python, SQL, machine learning, Tableau and Power BI through a project led curriculum.
- Mentored **NLP** and **computer vision** capstones end to end, from feature engineering through model evaluation to deployment.

PROJECTS

Inventory MCP Server on AWS
Nov 2026

Python, MCP SDK, AWS Lambda, API Gateway, SQLAlchemy, JSON Schema

- **Model Context Protocol** server on **AWS Lambda** behind API Gateway, exposing a warehouse database as **typed read only tools** rather than free form SQL, with SQLAlchemy backed handlers returning structured payloads.

Automated Product Image Pipeline
Oct 2026

n8n, Gemini, Gmail, Shopify Admin API, Python

- Built an event driven **n8n** workflow triggered on inbound mail matching a subject filter, routing every attached document into an agent step with no manual handling.
- Extracts embedded images from each document, enhances them through **Gemini**, then writes the assets into **Shopify** resolved to the correct product by its unique code.

Medha Voice Assistant
Feb 2026

Python, FastAPI, Sarvam AI, Claude API, LangChain, WebSockets

- Engineered a browser based **voice agent** with streaming speech to text over a WebSocket channel, speech synthesis on the return path, and **Claude** driving the reasoning.
- Held per session conversation state for multi turn context, with a **LangChain** agent loop selecting between web search and Wikipedia retrieval at run time.

Sakha, Sanskrit Motivational RAG Chatbot
Oct 2025

Python, ChromaDB, Sentence Transformers, Groq, Streamlit

- Built a **retrieval augmented chatbot** over a **ChromaDB** vector store, embedding a Sanskrit shloka corpus with Sentence Transformers for semantic retrieval.
- Added a **sentiment classification** stage conditioning both which passages are retrieved and how the response is framed, with generation served by **Groq**.

Conversational AI for SQL Inventory Search
May 2025

Python, SQLAlchemy, Fuzzy Matching, NLP

- Built a **natural language interface** classifying query intent and resolving entity names against the live schema with **fuzzy string matching**.
- Compiles to SQL through the **SQLAlchemy ORM**, keeping queries parameterised and the mapping resilient to schema changes; running in production at K.B. Ajmera.

CERTIFICATIONS

Large Language Models, Google, 2025
Prompt Design in Vertex AI, Google, 2025
Data Visualization Processing, IBM
Python Full Course, CDAC, Grade A
Cybersecurity Fundamentals, Internship

EDUCATION

B.Sc. Computer Science
Mumbai University, 2020 to 2023
GPA 9.9 / 9.6 / 9.4 (semester wise)